ASSIGNMENT 1

SCHOOL OF SCIENCE & TECHNOLOGY

SEMESTER:	**SUMMER SEMESTER 2026**
COURSE:	**STA 2030-A: STATISTICAL INFERENCE**
TIME :	**3 days**
INSTRUCTOR :	**JOYCE KIARIE, P.hD**
MODE:	**Group Assignment**
TOTAL MARKS :	**20 Marks**

Instructions

i. *For every applied part, **attach your R code**, the output, and a brief interpretation under the question.*
ii. *Any identical code or dataset submissions will be disqualified.*
iii. *Submit your groupwork as an individual in pdf format*

Recommended Real-World Data Sources (Kenya-Focused)

- Kenya Health Information System (KHIS) public indicators
- World Bank Data (Health, GDP, Population)
- Kenya Open Data Portal
- WHO TB data repository

Assignment Tasks

Part A: Data Understanding (4 Marks)

i. Describe dataset, variables, population of interest, and sampling unit.
ii. Explain relevance of dataset to a real decision problem.

Part B: Descriptive Statistics & Estimation (4 Marks)

i. Using R, compute mean, variance, SD, median and interpret in context.
ii. Include R code and outputs.

Part C: Confidence Interval (4 Marks)

i. Construct 95% CI for mean or proportion.
ii. Interpret findings for decision-making.

Part D: Hypothesis Testing (4 Marks)

i. Formulate H0 and H1 relevant to real-world question.
ii. Perform test, report p-value, and conclusion.

Part E: Bootstrap / Simulation (4 Marks)

i. Apply bootstrap resampling to estimate sampling distribution of the mean.
ii. Compare bootstrap CI with analytical CI.

Analytical Requirements

i. Each group must use a UNIQUE dataset or variable.
ii. Include raw downloaded dataset as appendix.
iii. Provide interpretation linked to a Kenyan policy or planning decision.
iv. Presentation slides will be required for class presentation

Expected Deliverables

1. Report (PDF)
2. R Script
3. Graphs (Histogram, Boxplot, Bootstrap Distribution)

Detailed Marking Scheme (20 Marks)

Component	Marks
Data understanding & relevance	4
Descriptive statistics & interpretation	4
Confidence interval computation & meaning	4
Hypothesis testing correctness & conclusion	4
Bootstrap implementation & comparison	4